Prodduturu Likith Kumar

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AI & ML Student passionate about building real-world Generative AI applications, RAG pipelines, and IoT automation.

Education

Madanapalle Institute of Technology and Science — Madanapalle, AP
*B.Tech in Artificial Intelligence & Machine Learning — **GPA: 7.3/10.0*** — *2024 – Present*

Loyola Polytechnic — YSSR, AP
*Diploma in Engineering — **Percentage: 83%*** — *Completed*

Technical Skills

Languages: Python, JavaScript, Embedded C
AI Frameworks: TensorFlow, PyTorch, YOLO, Langchain
Web Dev: Flask, Streamlit, HTML/CSS
Tools: Docker, Git, n8n, Arduino IDE
Databases: Pinecone (Vector DB), SQL
Platforms: Linux (Kali), VS Code, Google Colab

Featured Projects

Intelligent RAG-Based Chatbot | *GenAI, n8n, Pinecone* — December 2025
- Designed a campus support chatbot using **Retrieval-Augmented Generation (RAG)** to automate student inquiries.
- Orchestrated complex workflows using **n8n** and stored embeddings in a **Pinecone** vector database.
- Integrated the **Gemini 2.5 API** for high-accuracy, context-aware responses.
- View Repository

Fine-Tuned GPT-2 Medical Assistant | *LLM, NLP, PyTorch* — November 2025
- Fine-tuned a **GPT-2 model** on a dataset of 47,000+ medical Q&A pairs from NIH sources.
- Leveraged **QLoRA** and **PEFT** techniques to efficiently fine-tune model weights on an **RTX 3050 GPU**.
- View Repository

AI Powered Health Monitoring System | *YOLO, Flask, Computer Vision*
- Built a disease detection system for stenosis and mouth cancer using **YOLO** object detection.
- Developed a **Flask** backend API to serve predictions to a web interface for doctors.
- View Repository

Heart Stroke Prediction | *Scikit-learn, Streamlit, Python*
- Developed a predictive ML model to analyze patient health reports and estimate stroke risk.
- Built an interactive web dashboard using **Streamlit** for visualization and user interaction.
- View Repository

Security Alert Alarm System | *IoT, Arduino, Embedded C*
- Designed a hardware-based theft protection system using **Arduino** microcontrollers.
- Programmed real-time sensor logic using **Embedded C** to trigger security alerts.

Achievements

AI/ML Hackathon: Secured **14th place out of 65 teams** as a solo participant, demonstrating rapid prototyping skills.